

Hospital Resource Utilization & Readmission Risk Analytics

Introduction

Healthcare organizations generate large volumes of patient data from admissions, diagnoses, treatments, procedures, and follow-up visits. Analysing this data helps hospitals understand patient behaviour, optimize resource allocation, and identify factors associated with hospital readmissions.

This project focuses on analysing hospital admission patterns, diagnosis-related readmissions, resource utilization, and patient risk segmentation using data analytics techniques.

The objective is to transform raw hospital data into meaningful insights using Excel for data preparation, SQL for analytical querying, and Power BI for interactive dashboard development.

Project Overview

- Analyse overall patient admission patterns
- Identify diagnoses associated with hospital readmissions
- Understand hospital resource utilization
- Identify high-risk patient groups based on healthcare usage patterns
- Create an interactive dashboard for healthcare insights

Business Problems

Healthcare organizations face challenges in managing patient admissions, reducing avoidable readmissions, and efficiently utilizing hospital resources. This project addresses the following business problems:

1. Understanding Patient Admission Patterns

- How many patients are admitted to the hospital?
- Which patient groups contribute most to hospital visits?
- What are the demographic patterns of admitted patients?

2. Reducing Hospital Readmissions

- Which diagnoses are responsible for frequent readmissions?
- What is the rate of patients returning within 30 days?
- Which patient groups show higher readmission patterns?

3. Improving Resource Utilization

- Which patients require more hospital resources?
- How does hospital stay duration impact resource usage?
- Which factors contribute to increased healthcare workload?

4. Identifying High-Risk Patients

- Which patients show patterns associated with higher risk?
- What factors are commonly observed among high-risk patients?
- Which patient groups require closer monitoring and preventive care?

Dataset Overview

Dataset Name

Diabetes 130-US Hospitals for Years 1999-2008

Source

Kaggle Data Repository

Dataset Description

The dataset contains hospital encounter records of diabetic patients collected from multiple hospitals in the United States.

Original Dataset Size

Example:

- Rows: 101,766 encounters
- Columns: 50+ attributes

Project Dataset Used

For analysis optimization:

- Sample size: 25,000 patient encounters
- Random sampling performed
- Maintained original data distribution

Data Dictionary Overview

Important columns used:

Column	Description
encounter_id	Unique hospital visits ID
patient_nbr	Unique patient identifier
age	Patient age group
gender	Patient gender
diag_1	Primary diagnosis
diag_2	Secondary diagnosis
diag_3	Additional diagnosis
time_in_hospital	Length of stay
num_lab_procedures	Number of lab tests
num_medications	Medication count
number_emergency	Previous emergency visits
number_inpatient	Previous inpatient visits
readmitted	Readmission status

Tools Used for Analysis

- MS Excel
- MySQL
- Power BI

Methodology & Results

Step 1: Data Cleaning and Preparation

- ❖ Handled missing values by replacing "?" with "**Unknown**".
- ❖ Removed duplicate records to maintain data accuracy.
- ❖ Standardized data types and formats.
- ❖ Created new columns:
 - **Age Group** for patient segmentation.
 - **Risk Category** to classify High Risk and Low Risk patients.
- ❖ Reduced dataset size for faster processing.
- ❖ Prepared clean data for SQL analysis and Power BI visualization.

Step 2: SQL-Based Data Analysis

After data cleaning and preparation, the dataset was imported into MySQL for analytical processing.

2.1. Total Admissions Analysis

Methodology: The total number of hospital encounters was analysed to understand the overall patient volume and hospital workload.

Result:

- The dataset contained 25,000 hospital admissions/encounters.

Insight:

- The total admission count provides an overview of patient volume and helps understand the scale of healthcare demand.

2.2. Average Hospital Stay Analysis

Methodology: The average duration of hospitalization was calculated to understand the typical length of patient stay.

Result:

- The average hospital stay was approximately **4.72 days**.

Insight:

- Patients require nearly 5 days of hospitalization on average, indicating significant usage of hospital beds and healthcare resources.

2.3. Count of Readmissions Analysis

Methodology: Patient readmission records were analysed to identify the number of patients returning to the hospital.

Result:

- Patients not readmitted: **13,550**
- Readmitted after 30 days: **8,526**
- Readmitted within 30 days: **2,924**

Insight:

- A considerable number of patients experienced repeat hospital visits, highlighting the need to understand factors contributing to readmission.

2.4. Readmission Rate Analysis

Methodology: The percentage of patients readmitted within 30 days was calculated compared with total admissions.

Result:

- The 30-day readmission rate was approximately **11.7%**.

Insight:

- A portion of patients return shortly after discharge, indicating opportunities for better follow-up care and preventive management.

2.5. Top Diagnosis Analysis

Methodology: Diagnosis categories were analysed based on the number of hospital admissions to identify the most common conditions.

Result:

Top diagnosis categories included:

- Diagnosis code **428**
- Diagnosis code **250**
- Diagnosis code **414**
- Diagnosis code **276**
- Diagnosis code **401**

Insight:

- A few diagnosis categories contribute significantly to hospital visits and should be prioritized for healthcare planning.

2.6. Top Diagnosis Causing Readmissions Analysis

Methodology: Diagnosis categories were analysed specifically among patients who were readmitted within 30 days.

Result:

Top diagnoses associated with readmissions:

Diagnosis Readmissions	
428	615
250	411
414	400
276	364
401	312

Insight:

- Certain diagnosis groups have a stronger association with repeated hospital visits, indicating the need for targeted monitoring and follow-up programs.

2.7. Readmission by Age Analysis

Methodology: Readmission patterns were compared across different patient age groups.

Result:

- Older age groups showed higher readmission counts compared with younger groups.

Insight:

- Age is an important factor influencing healthcare utilization, with elderly patients requiring more continuous care.

2.8. Resource Utilization Analysis

Methodology: Healthcare resource usage was analysed using factors such as:

- Hospital stay duration
- Medication usage
- Emergency visits
- Previous inpatient visits

Result:

- Patients with higher healthcare utilization patterns showed increased resource requirements.

Insight:

- Patients with repeated hospital visits and longer stays contribute more to hospital resource consumption.

2.9. Diagnosis Distribution Analysis

Methodology: The overall distribution of diagnosis categories was analysed to understand disease patterns among admitted patients.

Result:

- Multiple diagnosis categories contributed to hospital admissions, with some conditions appearing more frequently.

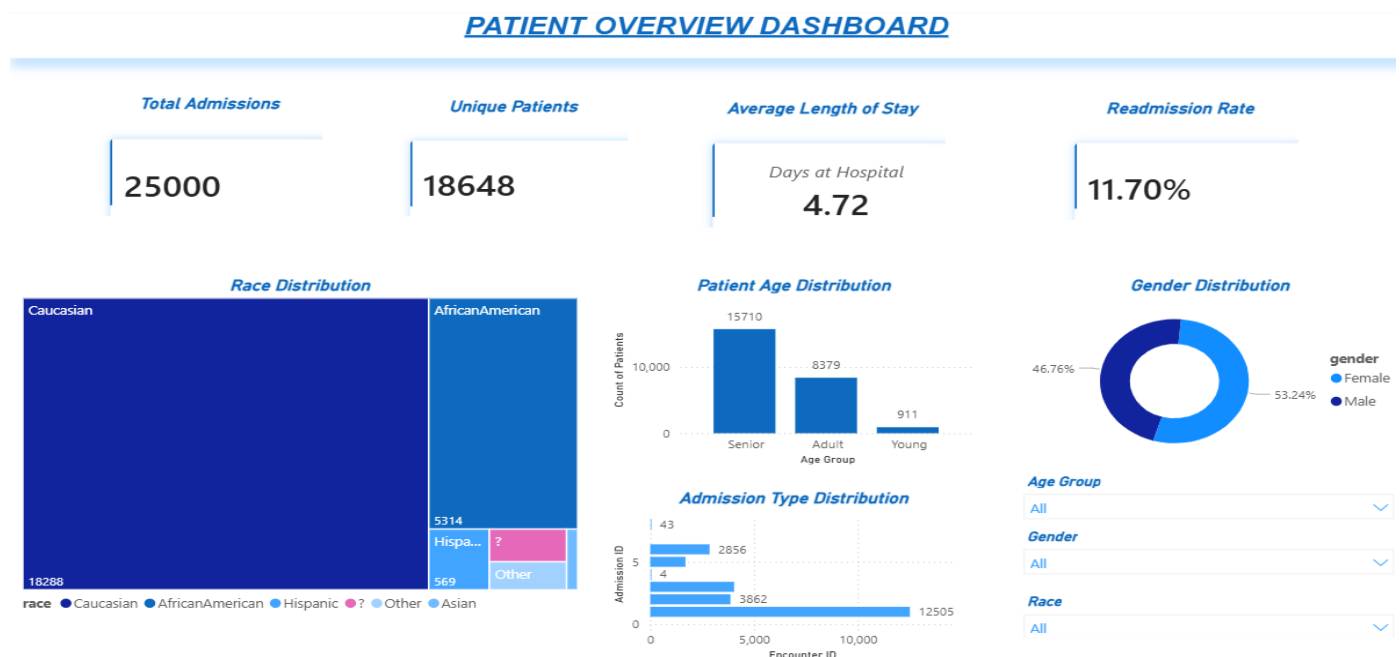
Insight:

- Understanding diagnosis distribution helps identify major disease areas and supports better resource planning.

Step 3: Visualizations

Dashboard 1: Patient Overview Dashboard

Objective: Hospital management needs a clear understanding of patient demographics, admission trends, and overall hospital utilization.



Key Results

Patient Population

- Total Admissions: **25,000**
- Unique Patients: **18,648**
- Readmission Rate: **11.7%**
- Average Length of Stay (LOS): **4.72 days**

Demographic Analysis

- Majority of patients belong to the **Senior Age Group (15,710 patients)**.
- Adult patients account for **8,379 admissions**.

- Young patients represent only **911 admissions**.

Gender Distribution

- Female: **53.24%**
- Male: **46.76%**

Race Distribution

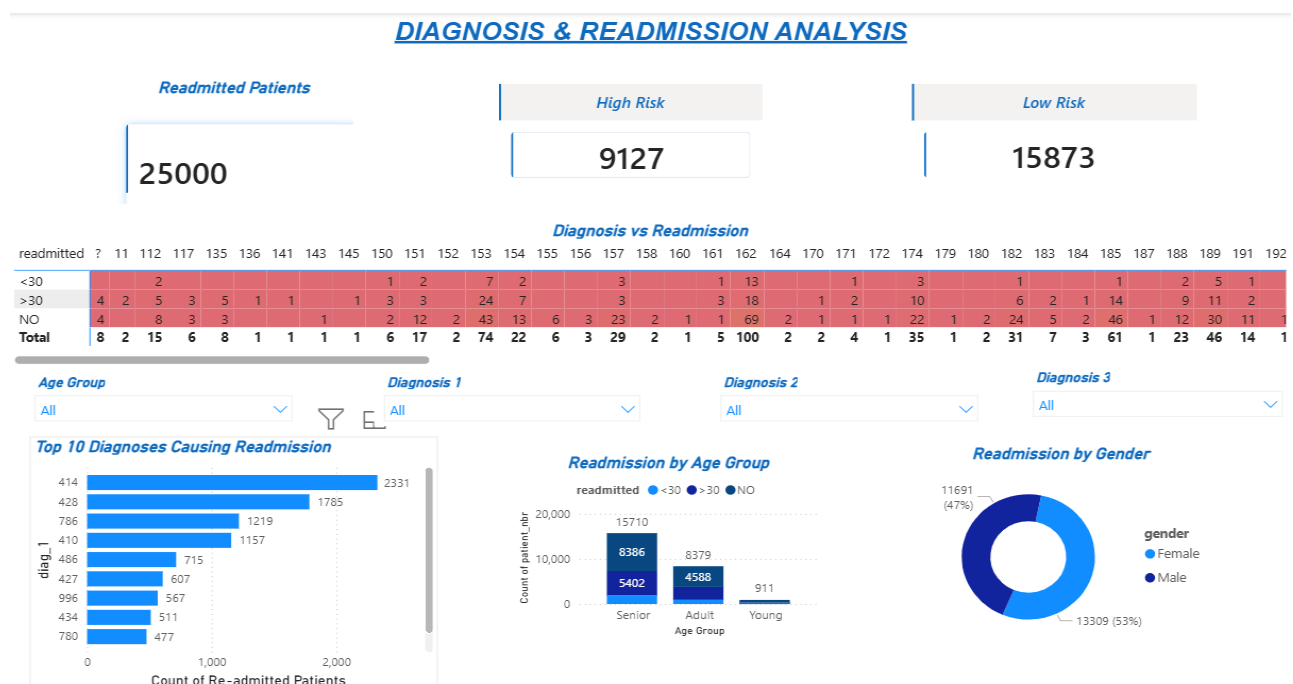
- Caucasian patients account for the largest proportion of admissions.
- African American patients form the second-largest patient group.

Business Insight

- Senior citizens represent the largest consumer of healthcare services.
- The hospital should prioritize geriatric care programs and specialized services for elderly patients.
- Readmission rate of 11.7% indicates room for improvement in discharge planning and follow-up care.

Dashboard 2: Diagnosis & Readmission Analysis

Objective: Which diagnoses contribute most to patient readmissions and which patient groups are at higher risk?



Key Results

Readmission Risk Categories

- High-Risk Patients: **9,127**
- Low-Risk Patients: **15,873**

Top Diagnoses Driving Readmissions

Most frequent readmission diagnoses include:

- Diagnosis 414
- Diagnosis 428
- Diagnosis 786
- Diagnosis 410
- Diagnosis 486

Age-Based Readmissions

- Senior patients have the highest readmission count.
- Adult patients have moderate readmission rates.
- Young patients contribute very little to readmissions.

Gender-Based Readmissions

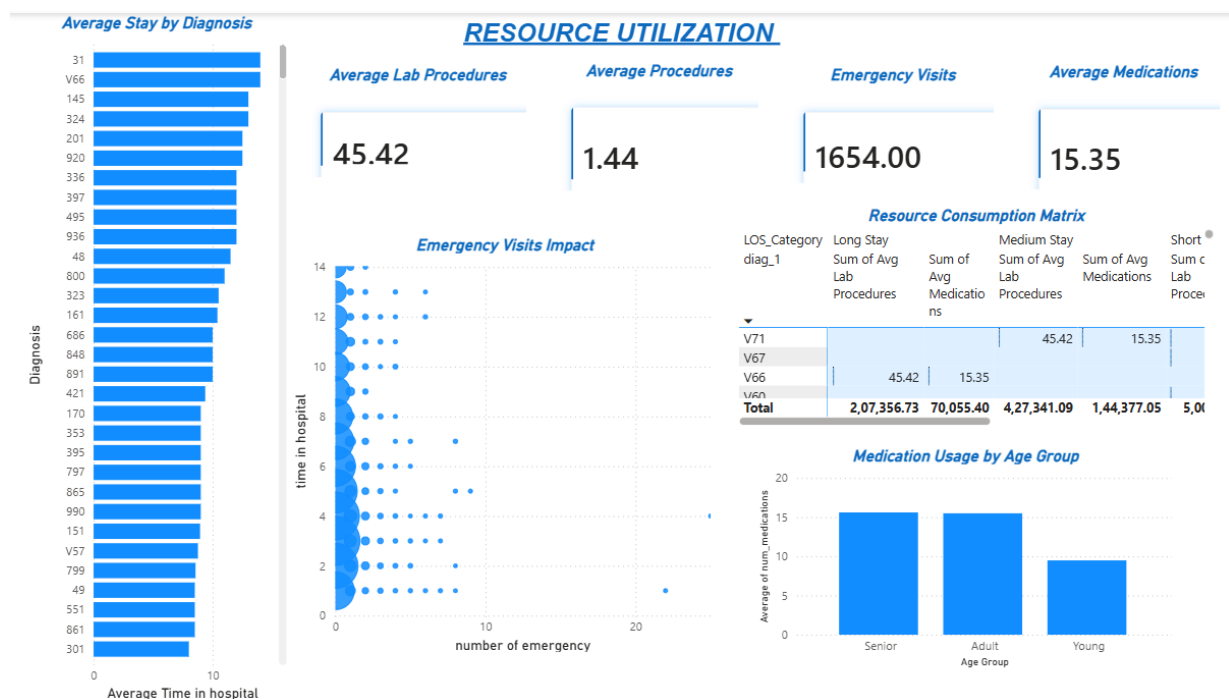
- Female patients: **53%**
- Male patients: **47%**

Business Insight

- Elderly patients are significantly more likely to be readmitted.
- Certain diagnoses repeatedly contribute to readmissions, suggesting chronic disease management challenges.
- Targeted care pathways for diagnosis codes 414 and 428 could substantially reduce readmission burden.

Dashboard 3: Resource Utilization

Objective: How efficiently are hospital resources being utilized and which patient groups consume the most resources?



Key Results

Resource Metrics

- Average Lab Procedures: **45.42**
- Average Procedures: **1.44**
- Average Medications: **15.35**
- Emergency Visits: **1,654** in total

Average Stay by Diagnosis

- Several diagnoses show significantly longer average stays compared to others.

Medication Usage by Age Group

- Seniors consume the highest medication volume.
- Adults have moderate medication usage.
- Young patients require fewer medications.

Emergency Visit Impact

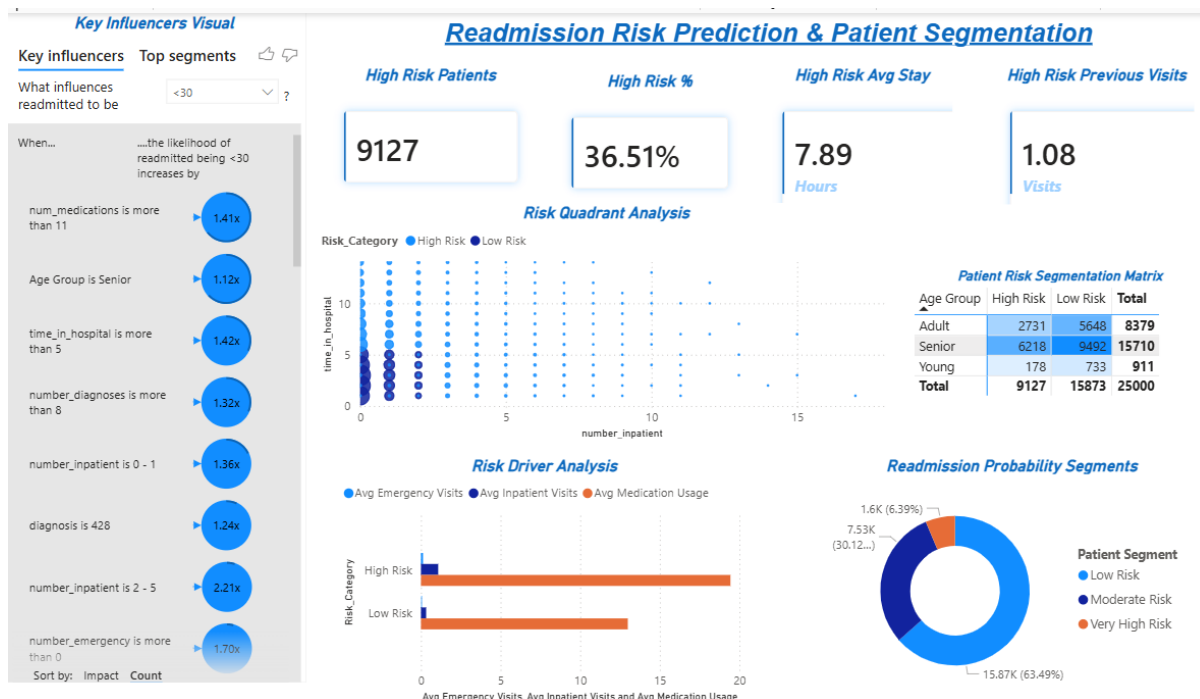
- Patients with higher emergency visits tend to have longer hospital stays.

Business Insight

- High emergency department utilization is associated with increased resource consumption.
- Senior patients require more medications and healthcare interventions.
- Certain diagnoses drive longer stays, increasing operational costs.

Dashboard 4: Readmission Risk Prediction & Patient Segmentation

Objective: Can hospitals proactively identify patients who are likely to be readmitted?



Key Results

High-Risk Population

- High-Risk Patients: **9,127**
- High-Risk Percentage: **36.51%**

High-Risk Patient Characteristics

Average:

- Length of Stay: **7.89 days**
- Previous Visits: **1.08**

Risk Segmentation

Segment	Patients
Low Risk	15,870
Moderate Risk	7,530
Very High Risk	1,600

Key Influencers of Readmission

The likelihood of readmission increases when:

1. Number of medications > 11
2. Age group = Senior
3. Hospital stay > 5 days
4. Number of diagnoses > 8
5. Diagnosis code = 428
6. Multiple inpatient visits
7. Higher emergency visits

Business Insight

- Medication burden is the strongest predictor of readmission.
- Seniors with multiple diagnoses and long hospital stays are the most vulnerable population.
- Diagnosis 428 appears consistently among high-risk readmitted patients and should be prioritized for intervention.

Conclusion

This project demonstrates how healthcare data can be transformed into actionable insights using **Excel, SQL, and Power BI**. The analysis provided a better understanding of patient admission patterns, diagnosis trends, hospital resource utilization, and readmission risk factors.

The analysis of **25,000 hospital admissions** identified an overall **11.7% 30-day readmission rate**, highlighting the importance of monitoring patients with repeated hospital visits. Senior patients represented a major patient segment and showed higher healthcare utilization patterns. Key factors associated with increased readmission risk included higher medication usage, longer hospital stays, multiple diagnoses, and previous emergency or inpatient visits.

The developed dashboards help identify high-risk patient groups, understand resource consumption patterns, and support data-driven healthcare decision-making. By focusing on early identification of high-risk patients and improving follow-up care strategies, healthcare organizations can work towards reducing avoidable readmissions, optimizing resource allocation, and improving patient outcomes.

References

1. Kaggle Dataset Repository - *Diabetes 130-US Hospitals Dataset*
2. Microsoft Excel
3. Microsoft Power BI Documentation
4. MySQL Documentation